

AXIS 0

Operator–SDF framework

A formal specification of Tom Klootwijk's definition

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Central definition

Axis 0 denotes the complete domain of operators named or generated within this framework. Every such operator is itself an operator signed-distance field. Bifurcation is expressed through Klein-bottle field geometry, with bit-shifted tree traversal, one-bit jitter, parity tracking, PSI evolution and lowercase φ scaling.

The specification preserves the author's final meaning of "0 = ALL operators" and turns the associated vocabulary into a mathematical interface. Source definitions, equations introduced for this formalization, and parameters awaiting an assigned interpretation remain distinguishable throughout.

SOURCE BASIS

The supplied six-page conversation and the identifiers supplied with this request. Section 11 records the source locations and the exact scope of the mathematical additions.

1 Definition and scope

SOURCE-DERIVED COMMITMENTS

1.1 The meaning of Axis 0

The controlling source statement is: “axis 0 is all operators themselves defined as signed distance fields where 0 just means ALL operators themselves in the prompts and my formalization.” [S1, p. 5]

Let $\mathcal{A}_0$ denote that domain. The expression *Axis 0* is its name; the subscript is not a Cartesian direction, an eigenvector index, a zero-valued field, or a claim that the number zero equals a set.

$$\text{Axis 0} := \mathcal{A}_0 = \{\text{all operator-fields admitted by this framework}\}. \quad (1.1)$$

The word *all* ranges over the operators in the prompts and their formal closure, not over an unstated universe of every possible mathematical function.

1.2 Defining commitments

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- A1 Every admitted operator is an operator-SDF, including operators used to construct, interpret, traverse or transform other operators.
 - A2 Every bifurcation is expressed as a Klein-bottle SDF object. Section 3 makes the relationship between its field and its geometric carrier precise.
 - A3 Traversal uses a binary tree, bit shifts, a one-bit jitter input and a one-bit orientation/parity state.
 - A4 PSI evolution, lowercase φ , log-encoded polar lookup, kinematic calculus and eigenvector structure belong to the same operator domain.
 - A5 The original operator vocabulary is retained, including symbols whose arity or evaluation law was not supplied.
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[S1, pp. 1--2 and 5]

1.3 How to read the equations

Source commitments above restate the author’s definitions. **Formalization choices** below introduce explicit notation and one coherent mathematical realization. **Parameters** name choices the source does not settle. These are different kinds of statements: a selected coordinate model is not retroactively attributed to the author.

In particular, the original conversation asks whether this is an engine, kernel or self-replicating referential universal framework. Section 9 defines those operational properties instead of treating the terminology alone as a demonstration of them. [S1, pp. 3--4]

2 The operator-field domain

FORMALIZATION CHOICE: A CLOSED FIELD LANGUAGE

2.1 Fields and their operator interpretation

Let $\mathcal{F}_\Theta$ be a class of field objects specified by a parameter collection Θ . Each field has a carrier space, a metric, a zero set and signed-distance data, together with a field-readable description that fixes its operator interpretation. The interpretation must distinguish operators with different behavior even when some of their geometry is shared.

For an ordinary chart, exact signed-distance data mean

$$|d_F(x)| = \text{dist}(x, Z_F), \quad Z_F = \{x : d_F(x) = 0\}. \quad (2.1)$$

For the Klein construction, the sign is chart-dependent and obeys the transition rule in Section 3. The absolute distance and the zero set are chart-independent. This is the precise *operator-SDF* convention adopted here.

Let Σ be the retained operator vocabulary and $\iota : \Sigma \rightarrow \mathcal{F}_\Theta$ its field realization. Let Γ be the collection of admitted field constructors, including composition and representation constructors. Define

$$A^{(0)} = \iota(\Sigma), \quad A^{(n+1)} = A^{(n)} \cup \Gamma(A^{(n)}), \quad \mathcal{A}_0 = \bigcup_{n \geq 0} A^{(n)}. \quad (2.2)$$

The notation $\Gamma(A)$ means all results of applying an admitted finite-arity constructor to operands in A . A finite description may contain reference links; self-reference need not be written as an infinitely expanded expression.

2.2 Closure and execution

Operator application is a field-valued partial operation:

$$\text{App} : \bigcup_{m \geq 0} (\mathcal{A}_0 \times \mathcal{A}_0^m) \rightarrow \mathcal{A}_0. \quad (2.3)$$

Whenever an application produces a result, that result is again an operator-field. A partial operation allows an expression to have no terminating evaluation; it does not assert that every expression terminates.

Numbers, bits, coordinates, tree records and states used by execution have field representations in $\mathcal{A}_0$. Their ordinary mathematical notation is used to describe those representations. Thus a bit written as 1 and the operator-field representing that bit are two levels of notation, not a second runtime universe outside Axis 0.

Internality condition

Every executable constructor, evaluator, lookup rule and transition rule has a representative in $\mathcal{A}_0$. The displayed mathematical maps describe how a field is read and evaluated; no particular geometry-to-instruction decoding is imposed without being included in Θ .

The source identifies operators with SDFs, but does not give that decoding map. Naming it explicitly preserves the identity requirement without inventing an instruction encoding. [S1, pp. 2 and 5]

3 Klein-bottle bifurcation fields

FORMALIZATION CHOICE: EXACT LOCAL DISTANCE WITH A TWISTED SIGN

3.1 Carrier and one-bit sign transition

Use coordinates $(u, v, z) \in \mathbb{R}^2 \times \mathbb{R}$ and the identifications generated by

$$A(u, v, z) = (u + 1, v, z), \quad B(u, v, z) = (-u, v + 1, -z). \quad (3.1)$$

Let $E = (\mathbb{R}^2 \times \mathbb{R}) / \langle A, B \rangle$ and let

$$K = \{[u, v, 0] \in E\}. \quad (3.2)$$

The zero section K has the Klein-bottle square-identification structure. Its two-sheeted orientation cover is a torus; this conventional topological background is recorded in [M1]. The extension in the z direction and the operator use of it are constructions introduced here.

Equip E with the quotient of the Euclidean metric. In any local lift define

$$d_i([u, v, z]) = z_i, \quad |d_i(x)| = \text{dist}_E(x, K). \quad (3.3)$$

The distance equality follows because a vertical segment reaches K with length $|z|$, while every path to K has length at least its change in $|z|$. Across an orientation-reversing transition,

$$d_j = -d_i, \quad p_j = p_i \oplus 1, \quad p_i \in \mathbb{B} := \{0, 1\}. \quad (3.4)$$

Thus the sign is locally carried, not assigned as a single global inside/outside label. The quantities $|d|$ and $d = 0$ are unambiguous on the quotient.

3.2 What “a bifurcation is a Klein bottle” means here

For each tree node w at which a split occurs, use a field object with neighborhood E_w and zero section $K_w \cong K$. Select two outgoing ports, labeled 0 and 1, and transition maps to the child fields. A port is an addressed location or chart, not a boundary component of the closed surface.

The tree indexes these field objects. Their local Klein geometry carries orientation-reversing loops; their two ports carry the branch choice. A binary graph and a surface therefore play distinct, explicitly connected roles within the same bifurcation object.

Bifurcation realization

$\text{Bif}(F)$ is an operator-field with Klein zero carrier K_F , exact local distance magnitude, a one-bit sign transition, and two addressed outgoing ports. Every such result belongs to $\mathcal{A}_0$.

This realizes the source’s Klein-bottle/SDF requirement without replacing it with an ordinary node bearing only a Klein-bottle label. [S1, pp. 1--2 and 5--6]

4 Tree traversal, jitter and parity

FORMALIZATION CHOICE: DISCRETE TRANSITION EQUATIONS

4.1 Logical address and bit shifting

Let $\mathcal{T} \subseteq \mathbb{B}^*$ be prefix-closed, with root ϵ . A node with both children has addresses $w0$ and $w1$. Let $q_n \in \mathbb{B}$ be the nominal branch decision and $j_n \in \mathbb{B}$ the supplied jitter bit. Define

$$c_n = q_n \oplus j_n, \quad w_{n+1} = w_n c_n. \quad (4.1)$$

Starting at integer address $a_0 = 1$, retain the full path by

$$a_{n+1} = (a_n \ll 1) | c_n = 2a_n + c_n. \quad (4.2)$$

Here $\ll 1$ is a one-place left shift and $|$ is bitwise OR. The leading 1 records the root, so leading zero branch choices are not lost. No finite machine-word width or maximum tree depth is assigned by these equations.

The jitter law is a parameter J that supplies j_n . It may be a specified deterministic stream or a specified stochastic process. “One bit” fixes the value set, not a probability distribution. [S1, pp. 1–2]

4.2 Orientation is a separate state

Let $h_n \in \mathbb{B}$ be the orientation reversal of the chosen chart/port transition. Then

$$p_{n+1} = p_n \oplus h_n, \quad d_{\text{new}} = (-1)^{h_n} d_{\text{old}} \quad (4.3)$$

for the same local geometric point expressed across that transition. The jitter changes a branch choice. The parity bit records orientation transport. They interact through the chosen route, but are not silently identified as the same bit.

For each local Klein carrier, an orientation-reversing closed route γ satisfies

$$\bigoplus_{e \in \gamma} h(e) = 1. \quad (4.4)$$

This distinguishes genuine twisted transport from an unrelated bit toggle on a tree. Taking the same reversing route twice restores the original parity.

4.3 The search-tree condition

To use the structure as an ordered binary search tree, provide a key map κ into a totally ordered key set. For every node w ,

$$\kappa(u) < \kappa(w) < \kappa(v) \quad (u \text{ in the left subtree, } v \text{ in the right subtree}). \quad (4.5)$$

An equal key selects the node rather than a child. Exact key lookup follows these comparisons. Jittered traversal describes exploration of the field tree; its branch selection alone does not certify that a requested key was found. A profile combining jitter with exact lookup must specify their relationship.

5 Lowercase phi, polar lookup and Cantor addresses

FORMALIZATION CHOICES: SCALING, COORDINATE ENCODING AND ADDRESS EMBEDDING

5.1 Why lowercase φ , not uppercase Φ

The source expressly requests lowercase phi. Here φ is the case-sensitive scale parameter; Φ is a different, unused symbol. The distinction follows this definition, not a universal rule assigning each Greek letter one exclusive mathematical meaning. [S1, p. 2]

Take $\varphi > 0$ for geometric scaling. No numerical value is fixed by the author's statements. The golden-ratio value $(1 + \sqrt{5})/2$ appears in the accompanying assistant explanation and is available as an additional choice, not silently inserted as an author-supplied constant. [S1, pp. 2--3]

$$r_{n+1} = \varphi^{\eta_n} r_n, \quad \eta_n \in \{-1, +1\}. \quad (5.1)$$

The exponent policy is a parameter. For $\varphi > 1$, $\eta_n = -1$ contracts and $\eta_n = +1$ expands.

5.2 Log-encoded polar LUT

On a selected two-dimensional local chart, let $r > 0$, $\theta \in [0, 2\pi)$, reference radius $r_* > 0$ and log base $b > 0$, $b \neq 1$. Define

$$\lambda = \log_b(r/r_*), \quad r = r_* b^\lambda, \quad \lambda_{n+1} = \lambda_n + \eta_n \log_b \varphi. \quad (5.2)$$

This defines “inverse logarithmically” as the inverse function b^λ , not the reciprocal $1/\log_b r$. It is an explicit realization of the source phrase. The origin uses the separate apex chart; it is not assigned a fabricated finite logarithm.

For $\Delta\lambda > 0$ and positive integer M , one possible LUT index is

$$i_r = \left\lfloor \frac{\lambda - \lambda_0}{\Delta\lambda} \right\rfloor, \quad i_\theta = \left\lfloor \frac{M\theta}{2\pi} \right\rfloor, \quad \text{LUT}(i_r, i_\theta) \in \mathcal{A}_0. \quad (5.3)$$

The stored values, key ordering and quantization parameters belong to Θ . A lookup table is not determined merely by its coordinate formula.

5.3 Cantor representation of branch paths

An infinite binary path $c_1 c_2 \dots$ is represented by

$$C(c) = \sum_{k=1}^{\infty} \frac{2c_k}{3^k}. \quad (5.4)$$

For a finite prefix of length n , set $C_n = \sum_{k=1}^n 2c_k/3^k$. All completions lie in $[C_n, C_n + 3^{-n}]$. The finite prefix is the stored address; the infinite series is its limiting address convention. This gives the source's Cantor-set vocabulary an explicit connection to tree traversal. [S1, p. 1]

6 Operator vocabulary and syntax

LITERAL SOURCE TERMS RETAINED; NO UNSTATED SYMBOL SUBSTITUTIONS

All entries below have operator-field representatives in $\mathcal{A}_0$. An entry designated a *parameter* keeps its name while its arity, operands and evaluation law remain to be assigned. The source does not supply those rules. [S1, pp. 1–2]

Source expression	Notation	Treatment in this specification
men T room	men T room	Literal label; parameter. No expansion of the phrase is assumed.
circle; sphere	Circle, Sphere	Geometric field constructors; chart and radius are operands.
cone; apex	Cone, Apex	Cone-field and distinguished apex data; dimensions and opening parameters are supplied by a profile.
legenda; side view of the pyramid	Legenda, SideView	Notation and view operators. Projection plane, pyramid geometry and output encoding are parameters.
PSI math +	Ψ , $+_{\text{PSI}}$	PSI names the evolution interface. The source-specific plus rule remains a parameter.
plus; encircled T	$+_{\text{src}}$, $T_{\circ}$	Kept distinct. A circled T is not replaced with XOR.
AT	AT	Literal operator label; parameter.
kindred search tree	Kindred	A tree-associated relation or selection operator, with its relation and rule supplied separately.
double UU set operator	UU	Family/set-like operator label; no automatic identification with ordinary union.
B floppy P	$B_{\text{floppy } P}$	Literal label; arity and rule unspecified. It is not replaced by a power-set, probability or elliptic-function symbol.
0 order computer science	ZeroOrder	Retained phrase; any operational meaning is a parameter. Axis 0 keeps the author's ALL-operators meaning.

6.1 Precedence and operator arity

For each label σ , let m_{σ} be its declared arity. Use fully parenthesized prefix expressions

$$e ::= \sigma \mid \sigma(e_1, \dots, e_{m_{\sigma}}). \quad (6.1)$$

This notation fixes grouping without inventing the source's missing infix precedence table. Infix rendering is permitted once a precedence and associativity table Π is specified. The XOR symbol $\oplus$ in Sections 3–4 denotes arithmetic on bits only; ordinary $+$ in numerical equations denotes addition. Neither assigns a meaning to $+_{\text{PSI}}$.

6.2 Geometric carriers and views

Geometric field constructors may have circle, sphere, cone or pyramid carriers. The Klein requirement applies to their *bifurcation field objects*; it does not require a circle itself to have Klein-bottle topology. A side view is a selected projection and is not assumed to be a unique reconstruction of its three-dimensional source.

7 Geometric transformation and kinematic calculus

FORMALIZATION CHOICES: EXACT DISTANCE SCALING AND LOCAL DYNAMICS

7.1 Exact positive scaling of a field

For a local SDF d_F , a positive scale s , an orthogonal matrix R and a translation t , define

$$d_{F'}(x) = s d_F(R^T(x - t)/s), \quad s > 0. \quad (7.1)$$

Its zero set is $Z_{F'} = t + sRZ_F$. To check the distance magnitude, write

$$\text{dist}(x, Z_{F'}) = \inf_{y \in Z_F} \|x - t - sRy\| \quad (7.2)$$

$$= s \inf_{y \in Z_F} \|R^T(x - t)/s - y\| = |d_{F'}(x)|. \quad (7.3)$$

The output factor s is therefore part of exact distance scaling. Choosing $s = \varphi^\eta$ connects the geometric transformation to Section 5. For the Klein fields, apply the rule in compatible local lifts and retain the sign-transition data.

7.2 Velocity, acceleration and eigenvectors

Let $x(t)$ be a twice-differentiable local coordinate trajectory of an operator-field. Define

$$v(t) = \frac{dx}{dt}, \quad a(t) = \frac{d^2x}{dt^2}. \quad (7.4)$$

A kinematic law is a specified field-represented function V_φ with $\dot{x} = V_\varphi(x, F, t)$. The source names kinematic vectors but supplies no velocity or acceleration law, so V_φ remains part of Θ . [S1, p. 2]

For the spatial part Ξ_φ of one discrete PSI step, on a differentiable chart with fixed branch choice, define

$$J_n = D_x \Xi_\varphi(x_n), \quad J_n e_{n,k} = \mu_{n,k} e_{n,k}. \quad (7.5)$$

This is the exact sense in which an eigenvector is used here. An eigenpair is supplied or computed for the actual local update. At a discrete branch transition, transport is given by the chart/port map instead of an assumed derivative across a discontinuity.

For example, the selected contraction $\Xi_\varphi(x) = x/\varphi$ has $J = \varphi^{-1}I$. Every nonzero vector is then an eigenvector with eigenvalue φ^{-1} . This conclusion concerns that particular map, not an unspecified PSI law.

Axis 0 is not an eigenvector axis

Differentiation, eigenvector extraction and coordinate transport are operators within $\mathcal{A}_0$. No preferred physical direction is substituted for the author's ALL-operators definition. [S1, p. 5]

8 Kernel transition and self-reference

FORMAL INTERFACE FOR AN OPERATIONAL REALIZATION

8.1 State and PSI step

Describe a state by

$$s_n = (w_n, a_n, x_n, p_n, F_n, M_n) \in \mathcal{S}, \quad (8.1)$$

where w_n is the logical node, a_n its integer address, x_n the local coordinates, p_n the parity, F_n the active field and M_n the tree/field store. Each state and each component used by execution has a representation in $\mathcal{A}_0$.

A parameter profile Θ supplies the nominal decision χ , port transport τ , orientation bit h , field update $\mathcal{U}$ and store update $\mathcal{W}$. One step is specified by

$$q_n = \chi(s_n), \quad c_n = q_n \oplus j_n, \quad (8.2)$$

$$x_{n+1} = \tau_{w_n, c_n, \varphi}(x_n), \quad p_{n+1} = p_n \oplus h(s_n, c_n), \quad (8.3)$$

$$F_{n+1} = \mathcal{U}(s_n, c_n, x_{n+1}, p_{n+1}), \quad M_{n+1} = \mathcal{W}(M_n, w_n c_n, F_{n+1}), \quad (8.4)$$

$$w_{n+1} = w_n c_n, \quad a_{n+1} = 2a_n + c_n. \quad (8.5)$$

Together these define $s_{n+1} = \Psi_\varphi(s_n, j_n)$ wherever the selected operations have an evaluation. The log-polar LUT may supply χ , $\mathcal{U}$ or both when the profile declares that connection.

The operational kernel is

$$\mathcal{K}_\Theta = (\mathcal{A}_0, \mathcal{S}, \Psi_\varphi). \quad (8.6)$$

Here “kernel” names the field language’s state-transition mechanism.

8.2 Field-readable self-reference

Let $Q(F) \in \mathcal{A}_0$ be a field-readable description of F , and let $\text{Read}(Q(F)) = F$. Introduce an evaluator field satisfying

$$\text{Eval}(Q(F), G) \simeq \text{App}(F; G). \quad (8.7)$$

The symbol $\simeq$ means the same terminating result, or nontermination on both sides. The description, reader and evaluator are themselves represented in $\mathcal{A}_0$.

For a field F_Ψ representing the step rule, the self-representation equation is

$$\text{Eval}(Q(F_\Psi), \text{Enc}(s, j)) \simeq \text{Enc}(\Psi_\varphi(s, j)). \quad (8.8)$$

This makes “referential” operational: the field language can represent and invoke its own transition rule. A geometric feedback loop is not substituted for the missing evaluator or description encoding.

9 Replication and universal execution

EXACT OPERATIONAL PROPERTIES FOR THE SOURCE'S CLASSIFICATION QUESTION

9.1 Self-replication

Let $D_\Theta \in \mathcal{A}_0$ describe a realized kernel, including its operator rules and required initial data. A replication operator realizes the contract

$$\text{Rep}(D_\Theta, M) \Downarrow (M', v_0, v_1), \quad v_0 \neq v_1, \quad (9.1)$$

where v_0, v_1 are fresh addresses and

$$\text{Read}(M'[v_0]) \equiv \text{Read}(D_\Theta) \equiv \text{Read}(M'[v_1]). \quad (9.2)$$

Here $\Downarrow$ denotes a terminating evaluation, and $\equiv$ means the same specified operator behavior. The two copies may use different geometric locations or φ scales when the reader preserves that behavior. The reproduction rule and the description that invokes it must both be included in the realized field language.

Two outgoing branches alone do not establish this equation. The source proposes self-replication in the discussion, but does not supply a replication transition or field-readable kernel description. This subsection gives the proposed property a precise contract. [S1, pp. 3–6]

9.2 Computational universality

Fix an effective encoding of Turing machines, and write $M_p(n)$ for the execution of the machine encoded by p on input n . A realization is computationally universal when there are computable encoding and decoding procedures E, D and a program field $U \in \mathcal{A}_0$ such that

$$D(\text{Run}_\Theta(U, E(p, n))) \simeq M_p(n) \quad \text{for every program } p \text{ and input } n. \quad (9.3)$$

The realization must specify a halting observation and output reading for Run_Θ . For a stochastic jitter law, the realization additionally states whether this equality is pathwise, almost sure, or replaced by a stated probability of correctness. A deterministic profile fixes the jitter stream or generator.

No concrete U, E, D or simulation proof is supplied in the source. Universality is therefore specified by the equation above, not inferred from the presence of a Klein carrier, an SDF, a Cantor address or a parity bit. [S1, pp. 3–4]

9.3 Names of realizations

An **engine** executes a selected family of field operations. The **kernel** is their state-transition mechanism. A **self-referential realization** satisfies Section 8's evaluator equation; a **self-replicating realization** satisfies the copy contract; a **universal realization** satisfies the simulation equation. These roles can coexist within one realization of Axis 0.

10 Worked trace and exact consequences

CALCULATED EXAMPLE OF THE SPECIFIED ADDRESS, PARITY AND SCALE RULES

Choose $a_0 = 1$, $p_0 = 0$, $r_0 = r_*$, $\varphi > 1$, log base $b = \varphi$ and contraction exponent $\eta_n = -1$. Supply the decision, jitter and orientation bits below. These are example inputs, not values inferred from an unspecified PSI evaluator.

Step	q	j	$c = q \oplus j$	h	New a	New p
1	0	1	1	1	3	1
2	1	0	1	0	7	1
3	1	1	0	1	14	0
4	0	0	0	1	28	1

The branch word is 1100. The stored address is $28 = (11100)_2$: removing the leading root bit recovers the branch word. The final parity is $0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 = 1$. The radius and log radius are

$$r_4 = r_* \varphi^{-4}, \quad \lambda_4 = -4. \quad (10.1)$$

The corresponding Cantor-prefix interval is

$$C_4 = \frac{2}{3} + \frac{2}{9} = \frac{8}{9}, \quad [C_4, C_4 + 3^{-4}] = \left[\frac{72}{81}, \frac{73}{81} \right]. \quad (10.2)$$

10.1 Consequences that follow from the equations

Recoverable addresses. Iterating $a_n = 2a_{n-1} + c_n$ gives

$$a_n = 2^n + \sum_{k=1}^n c_k 2^{n-k}. \quad (10.3)$$

The leading root bit and the n branch bits determine the address uniquely. Conversely, the least significant bit gives the last branch and shifting right gives its parent.

Parity composition. Iterating the XOR rule gives

$$p_n = p_0 \oplus \bigoplus_{k=1}^n h_k. \quad (10.4)$$

Two sign reversals cancel because $1 \oplus 1 = 0$. A sign reversal preserves the zero carrier and the distance magnitude because $|-d| = |d|$ and $d = 0$ exactly when $-d = 0$.

Field closure. Every constructor result belongs to the next level $A^{(n+1)}$ in Section 2. Consequently every finitely constructed output stays in Axis 0. These conclusions follow from the specified algebra; they do not depend on an unstated claim of universal execution.

11 Parameter register and source lineage

WHAT IS FIXED, WHAT IS SUPPLIED BY A REALIZATION, AND WHERE IT COMES FROM

11.1 Realization parameters

A concrete profile Θ assigns the following data. Free parameters are part of this specification's interface, not silently chosen constants.

Group	Required assignment
Field language	The field encoding ι , instruction reader, constructors Γ , quotation Q and evaluation rules.
Source labels	Arity and behavior of T_o , $+_{PSI}$, AT , UU , $B_{\text{floppy } P}$, men T room , Kindred and other retained labels.
Tree and topology	Key ordering, node/port locations, chart transports and associated parity map h .
Scale and lookup	φ , exponent policy η , b , r_* , λ_0 , $\Delta\lambda$, M , actual LUT entries and its search keys.
Evolution	The jitter law J , nominal decision χ , spatial transport τ , field update $\mathcal{U}$, store update $\mathcal{W}$ and any continuous kinematic law.
Execution properties	Halting/output rules; a replication rule for the replication contract; U , E , D and a simulation argument for universality.

11.2 Source record

[S1] Supplied six-page conversation. Opening text: “men T room a circle a sphere a cone an apex legenda a side view of the pyramid lower case phi log encoded polar LUT binary search tree klein bottle signed distance fields. . .” Page 1 contains the initial vocabulary; page 2 contains the author's extension with jitter, lowercase phi, “B floppy P”, kinematic vectors and bifurcation; page 3 contains the classification question; page 5 contains the final Axis 0 definition. The other passages are accompanying assistant explanations. Those explanations are not treated as proofs or as additional author-issued equations.

[S2] Current request. Tom Klootwijk; NL200678942; 10-07-1990; +31655954068. The date is reproduced as supplied, without assigning a document or identifier type beyond the request.

[M1] Mathematical background. Allen Hatcher, *Algebraic Topology*, Section 1.3, Example 1.42, printed page 74 (PDF page 83), [author-hosted edition](#). Used only for the standard Klein-bottle quotient and its two-sheeted torus cover. The extended signed-field realization, tree attachment and execution interfaces in this specification are introduced here, not attributed to that source.

11.3 Scope of this formalization

The author's domain definition and vocabulary are retained. The closure construction, twisted-sign distance model, recurrence equations, coordinate formulas and operational contracts make the definition mathematically expressible. The parameter register identifies the meanings still needed to select a particular executable realization.